

DYPAC-VLA: FULL-CONTEXT MIXED-PRECISION QUANTIZATION FOR VISION-LANGUAGE-ACTION MODELS

Anonymous authors

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ABSTRACT

Post-training quantization (PTQ) can reduce the static storage of vision-language-action (VLA) policies, yet errors that appear harmless at one layer can accumulate across action chunks and complete policy contexts. We present DYPAC-VLA—*Dynamic-Range and Prefix-Accumulated Control-aware Quantization*—a training-free mixed-precision framework that evaluates precision decisions in the complete policy. Its control metric, D_{PAC} , measures local action error, prefix accumulation, composed SE(3) pose drift, chunk-boundary discontinuity, and gripper-event timing, with a tail-risk penalty. Full-Context Precision Protection then searches native-FP16/group-64-W4 masks under an exact byte budget and accepts a change only after whole-network, cross-context adjudication. At deployment, Hessian-aware W4 weights are paired with DYRANGE-A8, which recomputes per-channel A8 scales on every forward pass; no task routing or runtime correction is used. On 50 RoboCasa365 tasks and 2,500 paired GR00T N1.5 episodes per configuration, DYPAC-VLA achieves **54.0%** success at $2.22\times$ static component compression. It improves over the previous GDSQ-VLA configuration by 3.2 points (Holm-adjusted $p = 0.0036$) and is statistically tied with the 55.1% FP16 teacher. A separate $\pi_{0.5}$ study reaches $2.702\times$ compression, but is reported only as a compression anchor because its 100-episode screen does not establish a success-rate gain.

1 INTRODUCTION

Vision-language-action (VLA) models connect visual perception and language-conditioned reasoning to continuous robot control Brohan et al. (2023); Octo Model Team et al. (2024); Kim et al. (2024); Black et al. (2024); Physical Intelligence et al. (2025); NVIDIA et al. (2025). Their large multimodal backbones and iterative action generators make static storage and memory bandwidth important deployment constraints. Post-training quantization (PTQ) is attractive because it reduces numerical precision without retraining the policy Xiao et al. (2023); Lin et al. (2024a).

Quantizing a VLA policy is not simply a layerwise reconstruction problem. An individually small action error can accumulate over a 16-step action chunk; rotation errors compose nonlinearly; discontinuities appear when consecutive replans are stitched together; and a small gripper timing shift can change task outcome. Moreover, the numerical effect of quantizing one layer depends on the precision of the rest of the policy. Local activation or reconstruction scores therefore need not predict the behavior of the complete mixed-precision network.

Activation range shift creates a second failure mode. Static A8 tables calibrated on a fixed buffer can become severely mismatched to the input distribution encountered by a generative controller. In our frozen attribution, the static-A8 reference has a normalized functional objective of 467, whereas dynamic A8 differs from the FP16 control by only 0.52. This gap motivates treating range adaptation as part of the deployment method, rather than as a calibration detail.

We propose DYPAC-VLA, short for *Dynamic-Range and Prefix-Accumulated Control-aware Quantization*. The central principle is to make precision decisions in the same context in which the policy will act. First, D_{PAC} compares paired FP16 and quantized action sequences in physical coordinates. It combines local and prefix-accumulated errors with SE(3) pose, inter-replan stitch, and gripper-event

terms, and adds CVaR tail risk. Second, Full-Context Precision Protection (FCP) uses counterfactual layer flips and an exact byte-constrained candidate generator, but lets a whole-network cross-context score make the final decision. Third, the selected W4/FP16 plan is deployed using group-64 Hessian-aware weights and DYRANGE-A8, a deterministic per-forward, per-input-channel A8 rule. The final method has one global static mask and uses neither task-specific routing nor runtime correction.

The full-context search also exposes an important negative result. On GR00T, none of 116 one-layer flips improves the paired proxy around the prior 100-W4/16-FP16 mask, and all six whole-network proposals violate the minimax or component-safety rule. Thus FCP validates that historical mask as a strict local optimum; it does not claim to discover a different mask. The final 3.2-point gain over GDSQ-VLA is obtained when the validated mask is run on the common Hessian-W4 + DYRANGE-A8 substrate.

We evaluate on RoboCasa365 [Nasiriany et al. \(2026\)](#) using 50 paired seeds on each of 50 tasks. On GR00T N1.5, DYPAC-VLA obtains 54.0% task-macro success at 962,068,480 static component bytes ($2.22\times$ compression), compared with 50.8% for the previous GDSQ-VLA runtime and 55.1% for FP16. The improvement over GDSQ-VLA is significant under paired McNemar testing with Holm correction ($p = 0.0036$); the difference from FP16 is not significant ($p = 0.2929$). On $\pi_{0.5}$, budget pruning increases the number of W4 layers from 80 to 121 and yields $2.702\times$ compression, but a 100-episode screen gives 29 versus 33 successes; we therefore make no success-rate superiority claim for that model.

Our contributions are threefold:

- (1) We introduce D_{PAC} , a control-space quantization metric that captures prefix accumulation, composed SE(3) drift, inter-replan stitching, gripper-event timing, and tail risk over complete action sequences.
- (2) We propose DYPAC-VLA, a training-free full-context precision architecture that combines exact-budget W4/FP16 search, whole-network cross-context adjudication, Hessian-aware group-64 W4, and per-forward DYRANGE-A8 without task routing or runtime correction.
- (3) We provide an artifact-audited evaluation showing 54.0% GR00T success: +3.2 points over GDSQ-VLA with Holm-adjusted $p = 0.0036$, while remaining statistically tied with FP16; we also report the $\pi_{0.5}$ compression boundary without overstating its success evidence.

2 RELATED WORK

Efficient VLA inference. Generalist VLA policies combine large multimodal encoders with continuous-control heads [Brohan et al. \(2023\)](#); [Octo Model Team et al. \(2024\)](#); [Kim et al. \(2024\)](#); [Black et al. \(2024\)](#); [Physical Intelligence et al. \(2025\)](#); [NVIDIA et al. \(2025\)](#). Efficiency methods use compact architectures, token reduction, temporal reuse, or architecture-quantizer co-design [Shukor et al. \(2025\)](#); [Zhang et al. \(2025\)](#); [Yang et al. \(2025\)](#); [Xu et al. \(2025\)](#); [Fang et al. \(2025\)](#). These approaches may reduce computation as well as storage, but alter the network or its execution. DYPAC-VLA instead preserves the pretrained checkpoint structure and studies static mixed-precision compression; we do not claim latency improvement for the eager fake-quant implementation.

VLA quantization. QuantVLA adapts distribution transformations to language and diffusion-action stacks [Zhang et al. \(2026\)](#). ActQuant allocates tensor precision from action contribution and action-aware curvature [Akbari et al. \(2026\)](#), while Ω -QVLA targets full-stack W4A4 with rotations and denoising-step scaling [Wang et al. \(2026\)](#). These methods show that control heads and time-varying distributions require specialized treatment. Our distinction is the decision context: DYPAC-VLA judges W4/FP16 changes through physical, prefix-accumulated action error in the complete policy, and deploys a single static mask without a correction selector.

Transformer and diffusion PTQ. GPTQ and AWQ use second-order information or activation salience for weight-only quantization [Frantar et al. \(2023\)](#); [Lin et al. \(2024b\)](#). Weight-activation PTQ reshapes difficult distributions through smoothing, calibration, and rotations [Xiao et al. \(2023\)](#); [Shao et al. \(2024\)](#); [Ashkboos et al. \(2024\)](#); [Lin et al. \(2024a\)](#); [Liu et al. \(2025\)](#); [Sun et al. \(2025\)](#). Diffusion PTQ further addresses timestep-dependent ranges and iterative error accumulation [Shang et al. \(2023\)](#); [Li et al. \(2023\)](#); [Wu et al. \(2024\)](#); [Chen et al. \(2025\)](#); [Zhao et al. \(2024\)](#). DYPAC-VLA builds

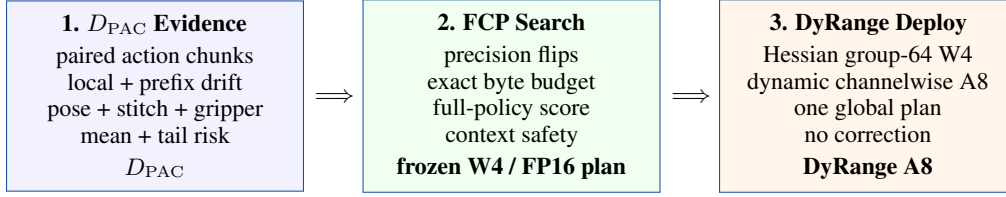


Figure 1: DYPAC-VLA architecture. Precision evidence is computed on paired physical action sequences, candidate masks are judged as complete policies under an exact byte budget, and the frozen plan is deployed with dynamic activation ranges. On GR00T, full-context adjudication retains the preceding 100-W4/16-FP16 mask rather than claiming a newly discovered allocation.

on a Hessian-aware group-64 W4 substrate. Its quantization novelty lies in full-context precision protection and dynamic per-forward activation ranges, not in a new weight-rounding primitive.

Mixed precision and behavioral sensitivity. Prior mixed-precision methods rank layers using hardware feedback, Hessian sensitivity, reconstruction loss, or diffusion-specific proxies Wang et al. (2019); Dong et al. (2019; 2020); Hubara et al. (2021); Li et al. (2021); Zhao et al. (2024). Representation similarities such as CKA and distribution divergences are useful diagnostics Kornblith et al. (2019); Yin et al. (2026), but a local feature discrepancy does not identify how errors accumulate in executed control. D_{PAC} instead operates on paired physical action sequences, and FCP reserves final authority for complete-network, multi-context evaluation under the deployment byte budget.

3 METHOD

Overview. DYPAC-VLA converts a frozen FP16 VLA into a single static W4/FP16 policy. Figure 1 summarizes three stages. First, paired teacher and candidate action chunks are compared with D_{PAC} . Second, FCP searches precision changes under an exact total-component byte budget and adjudicates complete candidates across task contexts. Third, the frozen plan is deployed with Hessian-aware group-64 W4 weights and dynamic per-forward A8 ranges. No success label, gradient, extra training data, task-dependent mask, runtime selector, or output correction is used.

3.1 PREFIX-ACCUMULATED CONTROL DIVERGENCE

Let $A^T = (a_1^T, \dots, a_H^T)$ and $A^Q = (a_1^Q, \dots, a_H^Q)$ be paired FP16-teacher and quantized action chunks generated from the same observation and initial action noise. We compare final physical actions after native inverse normalization, with $H = 16$ and translation, rotation, gripper, and other deployed coordinates. A robust per-coordinate penalty

$$\rho(z) = 2\left(\sqrt{1+z^2} - 1\right) \quad (1)$$

limits domination by isolated outliers. Its scale s is estimated once from the entire FP16 selection buffer using a MAD/RMS floor and is never re-estimated per sequence.

Local error alone ignores temporal accumulation. For action residual $e_h = a_h^Q - a_h^T$, the prefix term measures $\sum_{j=1}^h e_j / (s\sqrt{h})$ at every prefix. We additionally compose relative end-effector transforms in SE(3) for pose drift, compare the boundary between consecutive replans for stitch error, and measure both soft gripper state and open/close event timing. After normalizing the five components, a sequence score is

$$d_{PAC} = d_{\text{local}} + d_{\text{prefix}} + d_{\text{pose}} + d_{\text{stitch}} + d_{\text{grip}}. \quad (2)$$

The buffer-level divergence combines the sequence mean with $\text{CVaR}_{0.9}$, exposing rare catastrophic deviations that a mean can hide. We also retain a complementary functional diagnostic

$$D_{\text{func}} = d_{\text{final}} + d_{\text{kin}} + d_{\text{grip}} + 2 \text{CVaR}_{0.9}(d_{\text{solver}}), \quad (3)$$

but the complete selection rule must be safe under both D_{PAC} and D_{func} .

3.2 FULL-CONTEXT PRECISION PROTECTION

Context buffer. We collect FP16 on-policy observations from 12 tasks: four each from Atomic-Seen, Composite-Seen, and Composite-Unseen. Three seeds per task and four replans per sequence yield 36 sequences and 144 observations, stratified across early, middle, and late rollout windows. Task identifiers are used only to compute stratified statistics; the output remains one global mask.

Counterfactual search. Each target Linear layer has two states: native FP16 or signed-nibble group-64 W4. Starting from a complete mask, round one flips every W4 target to FP16 individually; round two flips every target state around the round-one winner. An exact sparse Pareto 0–1 dynamic program combines measured counterfactual benefits under

$$C(M) = C_{\text{fixed}} + \sum_i C_i(M_i) \leq 1.10 C_{\text{QuantVLA}}, \quad (4)$$

where C counts the same total static model components used in Table 1. Layer count is not a constraint. The dynamic program only proposes masks: non-additive interactions are decided after instantiating the complete network.

Whole-network adjudication. For every complete proposal, we recompute paired D_{PAC} and D_{func} deltas on all contexts. The scalar objective is the worst normalized one-standard-error upper bound over both metrics and the all-task, Atomic, Composite-Seen, and Composite-Unseen clusters. A candidate must make this bound strictly negative and separately satisfy non-positive upper bounds for pose, stitch, and gripper components. Candidate-state rescoring checks the top three plans on observations induced by the candidate itself. These conservative rules prevent a favorable average from masking a harmful context or control component.

On GR00T, all 116 one-layer flips have non-positive benefit relative to the preceding 100-W4/16-FP16 mask, and all six full-network proposals are ineligible. The frozen winner is therefore the unchanged context baseline. This result certifies a strict local optimum in the tested one-layer-flip neighborhood; it is not evidence that FCP discovered a new GR00T mask. For $\pi_{0.5}$, the budgeted plan flips the 41 lowest-benefit protected layers to W4, producing 121 W4 and 59 FP16 layers.

3.3 HESSIAN-W4 AND DYNAMIC-RANGE A8 DEPLOYMENT

For each W4 layer, 256 calibration observations form FP16 input second-order statistics $X^\top X$. A GPTQ-style sequential error-feedback pass selects clipping and quantizes signed weights in groups of 64 [Frantar et al. \(2023\)](#); row rotation is the identity and FP16 weights are not retained for W4 modules.

Static activation tables can fail under policy-induced range shift. DYRANGE-A8 instead recomputes a scale for every target Linear, forward call, and input channel:

$$s_c = \max\left(\frac{\max |x_{\dots,c}|}{127}, 10^{-6}\right), \quad q_c = \text{clip}(\text{round}(x_c/s_c), -127, 127). \quad (5)$$

The rule is deterministic, uses the input dtype, and has no calibration table, flow-step selector, task selector, or candidate branch. The final policy applies the frozen W4/FP16 mask and Eq. 5 uniformly to every input; runtime correction is disabled.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmark and models. We evaluate GR00T N1.5 and $\pi_{0.5}$ [Physical Intelligence et al. \(2025\)](#) on the RoboCasa365 v1.0.1 target split [Nasiriany et al. \(2026\)](#): 18 Atomic-Seen, 16 Composite-Seen, and 16 Composite-Unseen tasks. Formal GR00T configurations use 50 paired seeds per task, official horizons, a fresh rendered environment, four flow steps, and 16 executed actions per replan. Initial action noise is deterministically keyed by task, seed, and replan, producing 2,500 paired episodes per configuration.

Table 1: GR00T N1.5 success rate (SR, %) on RoboCasa365: 50 tasks, 50 paired seeds per task.

Configuration	W4 / FP16	Atomic	C-Seen	C-Unseen	All	GiB	Comp.
FP16 teacher	–	75.6	41.9	45.3	55.1	1.993	1.00×
QUANTVLA W4A8	116 / 0	49.6	19.6	19.8	30.4	0.898	2.22×
GDSQ-VLA (previous)	100 / 16	69.0	40.6	40.4	50.8	–	–
DYPAC-VLA (Ours)	100 / 16	74.7	43.9	40.9	54.0	0.896	2.22×

C-Seen/C-Unseen denote Composite-Seen/Unseen. FP16, QuantVLA, and ours use exact total static component accounting; the previous GDSQ-VLA row is retained as the paired runtime ablation, and its legacy storage scope is therefore omitted. Ours has 962,068,480 bytes.

Configurations and storage. FP16 is the teacher and QuantVLA W4A8 [Zhang et al. \(2026\)](#) is the uniform low-bit reference. We retain the previous GDSQ-VLA result to isolate the new common runtime. DYPAC-VLA uses 100 W4 and 16 native-FP16 target layers for GR00T, with no correction. Storage is exact total static model-component bytes, including fixed components and packing metadata but excluding activations, workspaces, simulator state, and fake-quant caches. The GR00T plan occupies 962,068,480 bytes, or 0.998× the QuantVLA cell and 2.22× compression relative to FP16.

Statistics and evidence gate. We report unweighted task-macro success, split macro success, exact paired two-sided McNemar tests, and a 10,000-draw task-then-seed hierarchical bootstrap. The three preregistered comparisons against GDSQ-VLA, FP16, and QuantVLA receive Holm correction. Frozen manifests bind plans, code, environments, tasks, seeds, and source hashes. A paper cell is generated only when exact coverage and artifact hashes pass the local audit.

4.2 GR00T MAIN RESULT

DYPAC-VLA obtains **54.0%** success, with 74.7%, 43.9%, and 40.9% on Atomic-Seen, Composite-Seen, and Composite-Unseen. It improves over the previous GDSQ-VLA runtime from 50.8% by 3.2 points. At the episode level there are 370 paired wins and 289 losses; the exact McNemar p -value is 0.0018 and the Holm-adjusted value is 0.0036. The hierarchical 95% interval is [0.48, 6.04] points, establishing formal superiority over that ablation.

FP16 obtains 55.1%. Ours has 292 wins and 319 losses, with $p = 0.2929$ and a 95% interval of [−3.88, 1.76] points. We therefore describe DYPAC-VLA as statistically tied with FP16, not equal to or better than it. QuantVLA obtains 30.4%; ours improves by 23.6 points with interval [19.12, 28.00] and Holm-adjusted $p < 10^{-4}$.

4.3 WHERE DOES THE GAIN COME FROM?

The full-context search returns the historical 100-W4/16-FP16 mask unchanged, so it cannot explain the 3.2-point gain by a new layer allocation. Instead, the controlled difference is the common deployment substrate: group-64 Hessian W4 and dynamic A8, with all corrections disabled. On five development tasks and ten new seeds, this runtime obtains 31 successes versus 25 for the historical runtime (seven paired wins, one loss). Separately, static A8 is strongly refuted by the offline paired objective (467), whereas DYRANGE-A8 is close to the FP16 activation control (0.52). These diagnostics support range adaptation as the principal implementation change, while FCP supplies evidence that changing the GR00T mask is unwarranted in the tested neighborhood.

This distinction is central to our claim: DYPAC-VLA is the complete validated mask plus the Hessian-W4/DYRANGE-A8 runtime. We do not attribute the GR00T gain to discovering a new mask, and the local-optimum result does not prove global optimality outside the searched counterfactual family.

4.4 $\pi_{0.5}$ COMPRESSION BOUNDARY

The $\pi_{0.5}$ plan increases W4 coverage from 80/180 to 121/180 layers, reducing exact static bytes to 0.886× the previous main plan and reaching 2.702× compression relative to FP16. In the combined 100-episode screen, however, the compressed candidate obtains 29 successes versus 33 for the prior

Table 2: $\pi_{0.5}$ compression anchor. The 100-episode screen is not a formal success comparison.

Configuration	W4	FP16	Success	Static GiB	Comp.
Previous main plan	80	100	33/100	1.718	2.394×
DYPAC-VLA anchor	121	59	29/100	1.523	2.702×

plan (seven paired wins, eleven losses). We register this result as a compression anchor and make no success-rate improvement, equivalence, or formal non-inferiority claim.

Deployment scope. The current implementation validates numerical behavior and exact packed storage accounting. Its eager fake-quant path retains floating-point execution, so the paper makes no end-to-end latency or live-memory claim. Fused low-bit kernels and real-robot evaluation remain future work.

5 CONCLUSION AND LIMITATIONS

We presented DYPAC-VLA, a training-free mixed-precision framework that evaluates quantization in complete policy context. D_{PAC} makes prefix accumulation, physical pose, action-chunk stitching, gripper timing, and tail risk visible to precision selection; FCP adjudicates complete masks under an exact byte budget; and DYRANGE-A8 adapts A8 scales on every forward call. The final deployment uses one static W4/FP16 mask, with no task routing or runtime correction.

On GR00T N1.5, DYPAC-VLA reaches 54.0% RoboCasa365 success at $2.22\times$ static component compression. It significantly improves over GDSQ-VLA by 3.2 points and is statistically tied with the 55.1% FP16 teacher. The result also clarifies mechanism: full-context search validates the existing GR00T mask as a local optimum, while the measured gain arises on the common Hessian-W4 + dynamic-A8 runtime.

Our evidence is bounded in three ways. The local search does not prove global mask optimality; the $\pi_{0.5}$ result is only a 100-episode compression anchor, not a success claim; and eager fake quantization does not establish latency or live-memory gains. Extending the full formal matrix to additional policy families, implementing fused INT4/INT8 kernels, and validating on real robots are the next steps.

AI USE STATEMENT

Generative AI tools assisted experiment planning, software implementation and debugging, result aggregation, drafting and editing, reference and table formatting, and architecture-figure preparation. The authors reviewed all generated material, tested AI-assisted code, reran evidence and document gates, verified numerical claims against frozen manifests, and checked references against primary sources. No experimental result was synthesized by a generative model. The authors take responsibility for the final content and artifacts.

REPRODUCIBILITY STATEMENT

Sections 3 and 4 specify the metric, full-context search, quantization, activation, evaluation, and statistical protocols. The supplementary artifact provides content-addressed plans, manifests, task/seed coverage, table generation, and claim–evidence audits. The headline numerical cells are regenerated from frozen summaries rather than entered manually.

REFERENCES

Arash Akbari, Arman Akbari, Masih Eskandar, Qitao Tan, Yixiao Chen, Jingwu Luo, Bertha Pangaribuan, Liyun Zhang, Jennifer Dy, Geng Yuan, Xue Lin, Gaowen Liu, Stratis Ioannidis, and Yanzhi Wang. Actquant: Sub-4-bit action-guided quantization for vision-language-action models. *arXiv preprint arXiv:2605.24011*, 2026. URL <https://arxiv.org/abs/2605.24011>.

- Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian Croci, Bo Li, Martin Jaggi, Dan Alistarh, Torsten Hoefler, and James Hensman. Quarot: Outlier-free 4-bit inference in rotated llms. In *Advances in Neural Information Processing Systems*, 2024.
- Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al. π_0 : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024. URL <https://arxiv.org/abs/2410.24164>.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *Conference on Robot Learning*, 2023. URL <https://arxiv.org/abs/2307.15818>.
- Lei Chen, Yuan Meng, Chen Tang, Xinzhu Ma, Jingyan Jiang, Xin Wang, Zhi Wang, and Wenwu Zhu. Q-dit: Accurate post-training quantization for diffusion transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 28306–28315, 2025. doi: 10.1109/CVPR52734.2025.02636.
- Zhen Dong, Zhewei Yao, Amir Gholami, Michael Mahoney, and Kurt Keutzer. Hawq: Hessian aware quantization of neural networks with mixed-precision. In *IEEE/CVF International Conference on Computer Vision*, 2019. URL <https://arxiv.org/abs/1905.03696>.
- Zhen Dong, Zhewei Yao, Yaohui Cai, Daiyaan Arfeen, Amir Gholami, Michael W. Mahoney, and Kurt Keutzer. Hawq-v2: Hessian aware trace-weighted quantization of neural networks. In *Advances in Neural Information Processing Systems*, 2020.
- Hengyu Fang, Yijiang Liu, Yuan Du, Li Du, and Huanrui Yang. Sqap-vla: A synergistic quantization-aware pruning framework for high-performance vision-language-action models. *arXiv preprint arXiv:2509.09090*, 2025. URL <https://arxiv.org/abs/2509.09090>.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefler, and Dan Alistarh. Gptq: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.17323>.
- Itay Hubara, Yury Nahshan, Yair Hanani, Ron Banner, and Daniel Soudry. Accurate post training quantization with small calibration sets. In *International Conference on Machine Learning*, volume 139, pp. 4466–4475. PMLR, 2021. URL <https://proceedings.mlr.press/v139/hubara21a.html>.
- Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Pannag Sanketi, Quan Vuong, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. URL <https://arxiv.org/abs/2406.09246>.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, 2019. URL <https://arxiv.org/abs/1905.00414>.
- Xiuyu Li, Yijiang Liu, Long Lian, Huanrui Yang, Zhen Dong, Daniel Kang, Shanghang Zhang, and Kurt Keutzer. Q-diffusion: Quantizing diffusion models. In *IEEE/CVF International Conference on Computer Vision*, pp. 17489–17499, 2023. doi: 10.1109/ICCV51070.2023.01608.
- Yuhang Li, Ruihao Gong, Xu Tan, Yang Yang, Peng Hu, Qi Zhang, Fengwei Yu, Wei Wang, and Shi Gu. Brcq: Pushing the limit of post-training quantization by block reconstruction. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2102.05426>.
- Haokun Lin, Haobo Xu, Yichen Wu, Jingzhi Cui, Yingtao Zhang, Linzhan Mou, Linqi Song, Zhenan Sun, and Ying Wei. Duquant: Distributing outliers via dual transformation makes stronger quantized llms. In *Advances in Neural Information Processing Systems*, 2024a.

- Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. Awq: Activation-aware weight quantization for llm compression and acceleration. *Proceedings of Machine Learning and Systems*, 6:87–100, 2024b.
- Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. Spinquant: Llm quantization with learned rotations. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2405.16406>.
- Soroush Nasiriany, Sepehr Nasiriany, Abhiram Maddukuri, and Yuke Zhu. Robocasa365: A large-scale simulation framework for training and benchmarking generalist robots. In *International Conference on Learning Representations*, 2026. URL <https://arxiv.org/abs/2603.04356>.
- NVIDIA, Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*, 2025. URL <https://arxiv.org/abs/2503.14734>.
- Octo Model Team, Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, et al. Octo: An open-source generalist robot policy. *arXiv preprint arXiv:2405.12213*, 2024. URL <https://arxiv.org/abs/2405.12213>.
- Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, and Ury Zhilinsky. $\pi_{0.5}$: A vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. doi: 10.48550/arXiv.2504.16054. URL <https://arxiv.org/abs/2504.16054>.
- Yuzhang Shang, Zhihang Yuan, Bin Xie, Bingzhe Wu, and Yan Yan. Post-training quantization on diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1972–1981, 2023. doi: 10.1109/CVPR52729.2023.00196.
- Wenqi Shao, Mengzhao Chen, Zhaoyang Zhang, Peng Xu, Lirui Zhao, Zhiqian Li, Kaipeng Zhang, Peng Gao, Yu Qiao, and Ping Luo. Omniquant: Omnidirectionally calibrated quantization for large language models. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2308.13137>.
- Mustafa Shukor, Dana Aubakirova, Francesco Capuano, Pepijn Kooijmans, Steven Palma, Adil Zouitine, Michel Aractingi, Caroline Pascal, Martino Russi, Andres Marafioti, et al. Smolvla: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025. URL <https://arxiv.org/abs/2506.01844>.
- Yuxuan Sun, Ruikang Liu, Haoli Bai, Han Bao, Kang Zhao, Yuening Li, Jiaxin Hu, Xianzhi Yu, Lu Hou, Chun Yuan, et al. Flatquant: Flatness matters for llm quantization. In *International Conference on Machine Learning*, 2025. URL <https://arxiv.org/abs/2410.09426>.
- Kuan Wang, Zhijian Liu, Yujun Lin, Ji Lin, and Song Han. Haq: Hardware-aware automated quantization with mixed precision. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- Xinyu Wang, Mingze Li, Sicheng Lyu, Dongxiu Liu, Kaicheng Yang, Ziyu Zhao, Yufei Cui, Xiao-Wen Chang, and Peng Lu. Omega-QVLA: Robust quantization for vision-language-action models via composite rotation and per-step scaling. *arXiv preprint arXiv:2605.28803*, 2026. URL <https://arxiv.org/abs/2605.28803v1>.
- Junyi Wu, Haoxuan Wang, Yuzhang Shang, Mubarak Shah, and Yan Yan. Ptq4dit: Post-training quantization for diffusion transformers. In *Advances in Neural Information Processing Systems*, volume 37, pp. 62732–62755, 2024. doi: 10.52202/079017-2006.

- Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. Smoothquant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning*, 2023.
- Siyu Xu, Yunke Wang, Chenghao Xia, Dihao Zhu, Tao Huang, and Chang Xu. Vla-cache: Efficient vision-language-action manipulation via adaptive token caching. In *Advances in Neural Information Processing Systems*, 2025. URL <https://arxiv.org/abs/2502.02175>.
- Yantai Yang, Yuhao Wang, Zichen Wen, Luo Zhongwei, Chang Zou, Zhipeng Zhang, Chuan Wen, and Linfeng Zhang. Efficientvla: Training-free acceleration and compression for vision-language-action models. *arXiv preprint arXiv:2506.10100*, 2025. URL <https://arxiv.org/abs/2506.10100>.
- Wenzhe Yin, Zehao Xiao, Pan Zhou, Shujian Yu, Jiayi Shen, Jan-Jakob Sonke, and Efstratios Gavves. Distributional vision-language alignment by cauchy–schwarz divergence. In *International Conference on Learning Representations*, 2026. URL <https://arxiv.org/abs/2502.17028>.
- Jingxuan Zhang, Yunta Hsieh, Zhongwei Wan, Haokun Lin, Xin Wang, Ziqi Wang, Yingtie Lei, and Mi Zhang. Quantvla: Scale-calibrated post-training quantization for vision-language-action models. *arXiv preprint arXiv:2602.20309*, 2026. URL <https://arxiv.org/abs/2602.20309>.
- Rongyu Zhang, Menghang Dong, Yuan Zhang, Liang Heng, Xiaowei Chi, Gaole Dai, Li Du, Yuan Du, and Shanghang Zhang. Mole-vla: Dynamic layer-skipping vision language action model via mixture-of-layers for efficient robot manipulation. *arXiv preprint arXiv:2503.20384*, 2025. URL <https://arxiv.org/abs/2503.20384>.
- Tianchen Zhao, Xuefei Ning, Tongcheng Fang, Enshu Liu, Guyue Huang, Zinan Lin, Shengen Yan, Guohao Dai, and Yu Wang. Mixdq: Memory-efficient few-step text-to-image diffusion models with metric-decoupled mixed precision quantization. In *European Conference on Computer Vision*, pp. 285–302, 2024. doi: 10.1007/978-3-031-72630-9_17.

A SUPPLEMENTARY EVIDENCE

A.1 FROZEN GR00T DECISION RECORD

The GR00T search starts from the previous 100-W4/16-FP16 mask. All 116 valid one-layer flips have non-positive benefit under the cross-context score. The exact dynamic program therefore emits no beneficial half- or full-budget combination, and all five additional structured proposals plus the context baseline are evaluated as complete networks. Every alternative has a positive paired minimax objective or violates pose, stitch, or gripper safety. The frozen fallback is the unchanged context baseline. These facts support local optimality only within the declared counterfactual neighborhood.

The final plan uses signed-nibble group-64 weights, identity row rotation, per-forward dynamic A8, and no ATM, OHB, error-folding, calibration bypass, or runtime selector. Its SHA-256 is recorded in the artifact registry and checked before table generation.

A.2 PAIRED STATISTICAL DETAIL

For ours versus GDSQ-VLA, 1,841 of 2,500 episodes are tied, 370 favor ours, and 289 favor GDSQ-VLA. Exact two-sided McNemar testing gives $p = 0.0018097$; Holm adjustment over the registered comparisons gives $p = 0.0036194$. The task-then-seed hierarchical bootstrap interval for the success-rate difference is $[0.0048, 0.0604]$. Against FP16, the corresponding counts are 1,889 ties, 292 wins, and 319 losses, with $p = 0.2929$ and interval $[-0.0388, 0.0176]$.

A.3 STORAGE ACCOUNTING

The GR00T FP16 static-component scope is 2,139,537,408 bytes. The final plan occupies 962,068,480 bytes, including 327,352,320 fixed bytes and packed mixed-precision target components, for $2.2239\times$ compression. This is $0.9982\times$ the QuantVLA cell of 963,772,416 bytes and satisfies the preregistered $1.10\times$ ceiling. The $\pi_{0.5}$ FP16 scope is 4,416,602,112 bytes; its 1,634,828,288-byte plan gives $2.7016\times$ compression.

These values are static packed estimates. Activations, CUDA workspaces, simulator state, and eager fake-quant caches are outside the scope; therefore no runtime speed or peak-memory conclusion follows from the compression ratios.

A.4 $\pi_{0.5}$ CLAIM GUARD

The $\pi_{0.5}$ combined screen uses two ten-seed waves over five tasks, for 100 paired episodes per configuration. The previous 80-W4 plan succeeds in 33 episodes and the 121-W4 candidate in 29; there are seven candidate wins and eleven losses. Since this screen neither advances to the 2,500-episode matrix nor passes a superiority gate, it supports only the observed storage point. The artifact labels the status `not_superior_to_gdsq_main`.